Chapter 6. Sequences and Series of Functions

Uniform convergence, differentiation and integration of limits, double series, power series, trigonometric series

Contents

6.1 Uniform convergence **2**

6.2 Differentiation and integration of limits **3**

6.3 Double sequences and double series **3**

6.4 Power series **5**

6.5 Trigonometric series **6**

6.6 Selected exercises **7**

6.1 Uniform convergence

Motivation. Pointwise limits can destroy continuity (x^n on $[0,1]$); uniform limits preserve it.

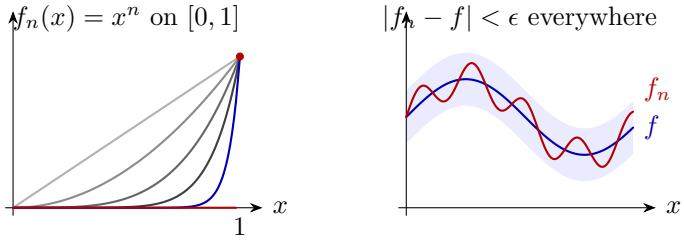


Figure. *Left: x^n has a discontinuous pointwise limit on $[0,1]$, so convergence is not uniform. Right: uniformly, the whole graph of f_n eventually lies in the ϵ-band around f.*

Let $X \subset \mathbb{R}^n$ and $f_n, f : X \to \mathbb{R}^m$ (or $\mathbb{C}$). $f_n \to f$ **pointwise** if $f_n(x) \to f(x)$ for each x, and **uniformly** if $\|f_n - f\|_{\sup} = \sup_{x\in X}\|f_n(x) - f(x)\| \to 0$.

Example 6.1

$x^n \to 0$ pointwise on $[0,1)$ but not uniformly, since $\sup_{[0,1)} x^n = 1$; the limit of x^n on $[0,1]$ is discontinuous. $f_n(x) = nx(1-x^2)^n$ tends to 0 pointwise on $[0,1]$, while $\int_0^1 f_n = \frac{n}{2n+2} \to \frac{1}{2}$.

Theorem 6.1 (6.1.1)

If continuous $f_n : X \to Y$ converge uniformly to f, then f is continuous.

Proof. Given x_0 and ϵ, choose N with $\|f_N - f\|_{\sup} < \epsilon/3$ and δ with $\|f_N(x) - f_N(x_0)\| < \epsilon/3$ for $\|x - x_0\| < \delta$. Then

$$\|f(x) - f(x_0)\| \le \|f(x) - f_N(x)\| + \|f_N(x) - f_N(x_0)\| + \|f_N(x_0) - f(x_0)\| < \epsilon.$$

■

Theorem 6.2 (6.1.2, uniform Cauchy criterion)

(f_n) converges uniformly on X iff for every $\epsilon > 0$ there is N with $\|f_m(x) - f_n(x)\| < \epsilon$ for all $m, n \ge N$ and $x \in X$.

Proof. Necessity is the triangle inequality. Conversely, each $(f_n(x))$ is Cauchy, so converges to some $f(x)$ (Theorem 2.14 of Volume I); letting $m \to \infty$ in $\|f_m(x) - f_n(x)\| < \epsilon$ gives $\|f(x) - f_n(x)\| \le \epsilon$ for all x and $n \ge N$. ■

Corollary 6.3 (6.1.3, Weierstrass M-test)

If $|f_n(x)| \le M_n$ on X and $\sum M_n < \infty$, then $\sum f_n$ converges absolutely and uniformly on X.

Proof. For $n > m$, $|\sum_{k=m+1}^{n} f_k(x)| \le \sum_{k=m+1}^{n} M_k$, which is small uniformly in x; apply Theorem 6.2 to the partial sums. ■

Example 6.2

$\sum 2^{-n}\cos(3^n x)$ converges uniformly, so it is continuous. Weierstrass's function $\sum a^n \cos(b^n \pi x)$ ($0 < a < 1$, $ab > 1 + 3\pi/2$, b odd) is continuous but nowhere differentiable. Also, there is a continuous curve filling a square (Peano), obtained as a uniform limit.

6.2 Differentiation and integration of limits

Motivation. Limits pass under integrals uniformly; under derivatives, the derivatives must converge uniformly.

Theorem 6.4 (6.2.1)

Let $f_n : [a, b] \to \mathbb{R}$ satisfy:
(a) each f_n is differentiable;
(b) $(f_n(x_0))$ converges for some $x_0 \in [a, b]$;
(c) (f_n') converges uniformly. Then (f_n) converges uniformly to a differentiable f, and $f'(x) = \lim f_n'(x)$.

Proof. By the mean value theorem applied to $f_m - f_n$,

$$|(f_m - f_n)(x) - (f_m - f_n)(t)| \le |x - t|\, \|f_m' - f_n'\|_{\sup}. \qquad (*)$$

With $t = x_0$: (f_n) is uniformly Cauchy, so $f_n \to f$ uniformly. Fix x; for $t \neq x$ let $\phi_n(t) = \frac{f_n(t) - f_n(x)}{t - x}$, $\phi(t) = \frac{f(t) - f(x)}{t - x}$. By $(*)$, $\phi_n \to \phi$ uniformly on $[a, b] - \{x\}$. With $g = \lim f_n'$, choose N with $|\phi_N - \phi| < \epsilon$ and $|f_N'(x) - g(x)| < \epsilon$, then δ with $|\phi_N(t) - f_N'(x)| < \epsilon$ for $0 < |t - x| < \delta$. Then $|\phi(t) - g(x)| < 3\epsilon$, so $f'(x) = g(x)$. ■

Uniform convergence of (f_n) alone is not enough: $\sin(nx)/\sqrt{n} \to 0$ uniformly, but the derivatives $\sqrt{n}\cos(nx)$ diverge.

Proposition 6.5 (6.2.2)

If Riemann integrable $f_n : [a, b] \to \mathbb{R}$ converge uniformly to f, then f is integrable and $\int_a^b f = \lim \int_a^b f_n$.

Proof. Let $\epsilon_n = \|f_n - f\|_{\sup}$. From $f_n - \epsilon_n \le f \le f_n + \epsilon_n$, the upper and lower integrals of f lie within $\epsilon_n(b - a)$ of $\int f_n$; so they differ by at most $2\epsilon_n(b - a) \to 0$, and $|\int f - \int f_n| \le \epsilon_n(b - a)$. ■

Example 6.3

Integrating $\frac{1}{1+t^2} = \sum_{n \ge 0}(-1)^n t^{2n}$ (uniform on $[-r, r]$, $r < 1$) gives $\arctan x = \sum (-1)^n x^{2n+1}/(2n + 1)$ for $|x| < 1$; similarly $\log(1 + x) = \sum (-1)^{n+1} x^n/n$. If $a_n > 0$ decrease and $\sum a_n < \infty$, then $na_n \to 0$: $\frac{n}{2} a_n \le a_{\lceil n/2 \rceil} + \cdots + a_n \to 0$.

6.3 Double sequences and double series

Motivation. Conditions, above all absolute convergence, under which iterated limits and sums commute.

Proposition 6.6 (6.3.1)

Let (a_{mn}) be a double sequence such that (a) $a_{mn} \to b_m$ as $n \to \infty$ for each m;
(b) $a_{mn} \to c_n$ as $m \to \infty$ for each n;
(c) the convergence in (b) is uniform in n: for every ϵ there is N with $|a_{mn} - c_n| < \epsilon$ for $m \geq N$ and all n. Then (b_m) and (c_n) converge to the same limit: $\lim_m \lim_n a_{mn} = \lim_n \lim_m a_{mn}$.

Proof. On $X = \{1/n\} \cup \{0\}$ let $f_m(1/n) = a_{mn}$, $f_m(0) = b_m$; f_m is continuous by (a), and uniformly convergent on X by (c) and Theorem 6.2 (at 0 since $|b_m - b_k| \leq \sup_n |a_{mn} - a_{kn}|$). The limit is continuous (Theorem 6.1) with values c_n at $1/n$ and $\lim b_m$ at 0, so $\lim_n c_n = \lim_m b_m$. ■

Proposition 6.7 (6.3.2)

If continuous $f_n : X \to Y$ converge uniformly to f and $x_n \to x$ in X, then $f_n(x_n) \to f(x)$.

Proof. $\|f_n(x_n) - f(x)\| \leq \|f_n - f\|_{\sup} + \|f(x_n) - f(x)\|$, and f is continuous (Theorem 6.1). ■

With pointwise convergence this fails: $f_n(x) = x^n$, $x_n = 1 - 1/n$, gives $f_n(x_n) \to 1/e$ while $f(1) = 1$.

Proposition 6.8 (6.3.3)

If $\sum_n |a_{mn}| = b_m < \infty$ for each m and $\sum_m b_m < \infty$, then $\sum_m \sum_n a_{mn} = \sum_n \sum_m a_{mn}$, all series converging absolutely.

Proof. Each $\sum_m |a_{mn}| \leq \sum_m b_m$, so the inner sums converge absolutely.

Let $S_{MN} = \sum_{m \leq M} \sum_{n \leq N} a_{mn}$. As $N \to \infty$, $S_{MN} \to \sum_{m \leq M} \sum_n a_{mn}$; as $M \to \infty$, $S_{MN} \to \sum_{n \leq N} \sum_m a_{mn}$, uniformly in N, since

$$\left| S_{MN} - \sum_{n \leq N} \sum_m a_{mn} \right| \leq \sum_{m > M} b_m.$$

Proposition 6.6 gives the equality of the iterated limits. ■

Proposition 6.9 (6.3.4)

If $a_{mn} \geq 0$ for all m, n, then $\sum_m \sum_n a_{mn} = \sum_n \sum_m a_{mn}$ in $[0, \infty]$.

Proof. For all M, N, $\sum_{m \leq M} \sum_{n \leq N} a_{mn} \leq \sum_{n \leq N} \sum_m a_{mn} \leq \sum_n \sum_m a_{mn}$. Letting $N \to \infty$ and then $M \to \infty$ gives $\leq$; the reverse is symmetric. ■

Both sides also equal $\sup_F \sum_{(m,n) \in F} a_{mn}$ over finite sets F, and any enumeration of the double series of nonnegative terms has this sum (Proposition 2.16 of Volume I). For $(a_k)_{k \in \mathbb{Z}}$, $\sum_{k=-\infty}^{\infty} a_k$ means $\lim_{m,n \to \infty} \sum_{k=-m}^{n} a_k$ when this double limit exists.

Corollary 6.10 (6.3.5)

If $a_k \geq 0$ for all $k \in \mathbb{Z}$, then $\lim_n \sum_{-n}^{n} a_k = \lim_n \lim_m \sum_{-m}^{n} a_k = \lim_m \lim_n \sum_{-m}^{n} a_k$ in $[0, \infty]$.

Proof. All three equal $\sup_{m,n} \sum_{-m}^{n} a_k$, since the partial sums increase in both m and n. ■

Corollary 6.11 (6.3.6)

If $\sum_{k \in \mathbb{Z}} |a_k| < \infty$, the three limits in Corollary 6.10 exist and coincide.

Proof. Apply Corollary 6.10 to $a_k^{\pm} = \max(\pm a_k, 0)$, whose sums are finite, and subtract. ∎

6.4 Power series

Motivation. Power series converge uniformly inside their radius and differentiate termwise, defining e^x, $\sin x$, $\cos x$.

Theorem 6.12 (6.4.1, Cauchy–Hadamard)

Let $\lambda = \limsup|a_n|^{1/n}$ and $R = 1/\lambda$ ($R = \infty$ if $\lambda = 0$, $R = 0$ if $\lambda = \infty$). Then $\sum a_n x^n$ converges absolutely for $|x| < R$ and diverges for $|x| > R$; for every $0 < r < R$ it converges uniformly on $[-r, r]$. Its sum f is differentiable on $(-R, R)$ with $f'(x) = \sum n a_n x^{n-1}$, and hence C^∞, with $a_n = f^{(n)}(0)/n!$.

Proof. $\limsup|a_n x^n|^{1/n} = \lambda|x|$; apply the root test (Proposition 2.19 of Volume I). On $[-r, r]$ use the M-test with $M_n = |a_n| r^n$, summable as $r < R$. Since $\limsup|n a_n|^{1/n} = \lambda$ ($n^{1/n} \to 1$), the derived series has the same radius and converges uniformly on $[-r, r]$; Theorem 6.4 gives $f' = \sum n a_n x^{n-1}$. Iterate. ∎

Theorem 6.13 (6.4.2, re-expansion)

If $f(x) = \sum a_n x^n$ on $(-R, R)$ and $|c| < R$, then $f(x) = \sum_n \frac{f^{(n)}(c)}{n!}(x-c)^n$ for $|x-c| < R - |c|$.

Proof. $f(x) = \sum_n a_n(c+(x-c))^n = \sum_n \sum_{k \le n} a_n \binom{n}{k} c^{n-k}(x-c)^k$. Since $\sum_n \sum_k |a_n| \binom{n}{k} |c|^{n-k}|x-c|^k = \sum_n |a_n|(|c| + |x-c|)^n < \infty$, Proposition 6.8 allows the order of summation to be exchanged: $f(x) = \sum_k (\sum_n \binom{n}{k} a_n c^{n-k})(x-c)^k$, and the coefficient of $(x-c)^k$ is $f^{(k)}(c)/k!$ by termwise differentiation (Theorem 6.12). ∎

So a power series is **analytic** (locally a power series) throughout its interval of convergence. The **Cauchy product** of $\sum a_n$ and $\sum b_n$ is $\sum c_n$, $c_n = \sum_{k=0}^{n} a_k b_{n-k}$.

Proposition 6.14 (6.4.3, Mertens)

If $\sum a_n = A$ converges absolutely and $\sum b_n = B$, then $\sum c_n = AB$.

Proof. With partial sums A_n, B_n, C_n and $\beta_n = B_n - B$, $C_n = \sum_{k \le n} a_k B_{n-k} = A_n B + \gamma_n$, $\gamma_n = \sum_{k \le n} a_k \beta_{n-k}$. With $\alpha = \sum|a_k|$ and $|\beta_m| < \epsilon$ for $m \ge N$, $|\gamma_n| \le \epsilon\alpha + \sum_{n-N < k \le n} |a_k||\beta_{n-k}|$, and the last sum $\to 0$ ($a_k \to 0$, β bounded). So $C_n \to AB$. ∎

Theorem 6.15 (6.4.4, Abel)

Let $f(x) = \sum a_n x^n$ on $(-R, R)$. If $\sum a_n R^n$ converges, then $\lim_{x \to R^-} f(x) = \sum a_n R^n$.

Proof. Take $R = 1$ ($a_n \to a_n R^n$) and $\sum a_n = 0$ (subtract a constant). With partial sums $s_n \to 0$, summation by parts gives $\sum_{n \le N} a_n x^n = (1-x)\sum_{n<N} s_n x^n + s_N x^N$, so $f(x) = (1-x)\sum s_n x^n$. If $|s_n| < \epsilon$ for $n \ge M$, then $|f(x)| \le (1-x)\sum_{n<M}|s_n| + \epsilon < 2\epsilon$ for x near 1^-. ∎

Corollary 6.16 (6.4.5)

If $\sum a_n = A$, $\sum b_n = B$ and their Cauchy product $\sum c_n = C$ all converge, then $C = AB$.

Proof. For $|x|<1$ the series $\sum a_n x^n$, $\sum b_n x^n$ converge absolutely, so $\sum c_n x^n = (\sum a_n x^n)(\sum b_n x^n)$ by Proposition 6.14. Let $x \to 1^-$ and use Theorem 6.15 three times. ∎

Example 6.4

Abel's theorem gives $\log 2 = 1 - \frac{1}{2} + \frac{1}{3} - \cdots$ and $\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \cdots$ from Example 6.3. Defining $e^x = \sum x^n/n!$, Proposition 6.14 gives $e^{x+y} = e^x e^y$; with $\sin, \cos$ defined by their series, $(\sin)' = \cos$, $(\cos)' = -\sin$, and $\sin^2 + \cos^2 = 1$ follow by termwise differentiation.

6.5 Trigonometric series

Motivation. Uniform convergence of trigonometric series, preparing Chapter 9.

A complex vector space V with a map $\langle \cdot, \cdot \rangle$ that is linear in the first variable, satisfies $\langle y, x\rangle = \overline{\langle x, y\rangle}$, and $\langle x, x\rangle > 0$ for $x \neq 0$, is an **inner product space**; $\|x\| = \langle x, x\rangle^{1/2}$.

Theorem 6.17 (6.5.1, Cauchy–Schwarz)

$|\langle x, y\rangle| \le \|x\|\|y\|$ in a complex inner product space; hence $\|x+y\| \le \|x\| + \|y\|$.

Proof. If $y \neq 0$, put $\lambda = \langle x, y\rangle / \|y\|^2$; then $0 \le \|x - \lambda y\|^2 = \|x\|^2 - |\langle x, y\rangle|^2/\|y\|^2$. Then $\|x+y\|^2 = \|x\|^2 + 2\,\mathrm{Re}\langle x, y\rangle + \|y\|^2 \le (\|x\| + \|y\|)^2$. ∎

Summation by parts: with $F_n = \sum_{k \le n} f_k$ ($F_0 = 0$),

$$\sum_{k=m}^{n} f_k g_k = F_n g_n - F_{m-1} g_m + \sum_{k=m}^{n-1} F_k (g_k - g_{k+1}). \tag{†}$$

Theorem 6.18 (6.5.2, Dirichlet–Dedekind test)

Let $f_n, g_n : X \to \mathbb{C}$ satisfy:
(a) the partial sums F_n of $\sum f_n$ are uniformly bounded, $|F_n(x)| \le K$;
(b) $g_n \to 0$ uniformly;
(c) $\sum_n |g_n(x) - g_{n+1}(x)|$ converges uniformly on X. Then $\sum f_n g_n$ converges uniformly on X.

Proof. By (†), $|\sum_{k=m}^{n} f_k g_k| \le K|g_n| + K|g_m| + K\sum_{k=m}^{n-1}|g_k - g_{k+1}|$, which is uniformly small for large m; apply Theorem 6.2. ∎

Corollary 6.19 (6.5.3)

If the partial sums of $\sum a_n$ are bounded and b_n decreases to 0, then $\sum a_n b_n$ converges.

Proof. Here $\sum |b_n - b_{n+1}| = b_1$; apply Theorem 6.18 with constant functions. ∎

This contains the alternating series test (Theorem 2.17 of Volume I). For $x \notin 2\pi\mathbb{Z}$,

$$\sum_{k=1}^{n} e^{ikx} = e^{ix}\frac{e^{inx} - 1}{e^{ix} - 1}, \qquad \Big|\sum_{k=1}^{n} e^{ikx}\Big| \le \frac{1}{|\sin(x/2)|}, \tag{‡}$$

so the partial sums of $\sum \sin kx$ and $\sum \cos kx$ are bounded by $1/\sin(\delta/2)$ on $[\delta, 2\pi - \delta]$.

Theorem 6.20 (6.5.4)

If $a_n \to 0$ and $\sum |a_n - a_{n-1}| < \infty$, then for every $\delta \in (0, \pi)$ the series $\sum a_n \sin nx$ and $\sum a_n \cos nx$ converge uniformly on $[\delta, 2\pi - \delta]$.

Proof. Theorem 6.18 with $f_n(x) = e^{inx}$ (bounded partial sums by (‡)) and constant $g_n = a_n$; take real and imaginary parts. ■

Theorem 6.21 (6.5.5)

Let a_n decrease to 0. Then $\sum a_n \sin nx$ converges uniformly on $\mathbb{R}$ iff $na_n \to 0$.

Proof. ($\Rightarrow$) For large n, $|\sum_{n/2 \le k \le n} a_k \sin kx| < \epsilon$ for all x. At $x = \pi/(2n)$, $kx \in [\pi/4, \pi/2]$ gives $\sin kx \ge 1/\sqrt{2}$, and $a_k \ge a_n$ for these $\ge n/2$ terms, so $\frac{n}{2} a_n/\sqrt{2} < \epsilon$.

($\Leftarrow$) (†), (‡) and $\sin(x/2) \ge x/\pi$ give $|\sum_{k=m}^{n} a_k \sin kx| \le 2a_m / \sin(x/2) \le 2\pi a_m / x$ on $(0, \pi]$. Let $\epsilon_m = \sup_{k \ge m} k a_k \to 0$, $R_m(x) = \sum_{k \ge m} a_k \sin kx$ (by oddness and periodicity, $x \in (0, \pi]$ suffices), and $N = \lfloor \pi/x \rfloor$. If $m > N$: $m > \pi/x$, so $|R_m(x)| \le 2\pi a_m/x < 2m a_m \le 2\epsilon_m$. If $m \le N$: $|\sum_{k=m}^{N}| \le N \cdot \epsilon_m x \le \pi \epsilon_m$ (as $|a_k \sin kx| \le k a_k x$), and $|\sum_{k>N}| \le 2\pi a_{N+1}/x < 2(N+1) a_{N+1} \le 2\epsilon_m$. So $\sup_x |R_m(x)| \le (\pi + 2)\epsilon_m \to 0$. ■

Theorem 6.22 (6.5.6, Abel's test)

Let $f_n : X \to \mathbb{C}$ and $g_n : X \to \mathbb{R}$ satisfy:
(a) $\sum f_n$ converges uniformly;
(b) $g_{n+1}(x) \le g_n(x)$ for all x, n;
(c) (g_n) is uniformly bounded, $|g_n| \le M$. Then $\sum f_n g_n$ converges uniformly.

Proof. By (†) with $R_k = \sum_{j=m}^{k} f_j$, $\sum_{k=m}^{n} f_k g_k = R_n g_n + \sum_{k=m}^{n-1} R_k (g_k - g_{k+1})$. For large m, $|R_k| < \epsilon$ ($k \ge m$) by (a), and $g_k - g_{k+1} \ge 0$, so the sum is $\le \epsilon M + \epsilon(g_m - g_n) \le 3M\epsilon$. ■

6.6 Selected exercises

Motivation. Book exercises: Dini's theorem, termwise integration, a conditional Cauchy product, and Tauber's theorem.

Exercise 6.6.6 (Dini's theorem)

Let $K \subset \mathbb{R}^n$ be compact and let $f_n : K \to [0, \infty)$ be continuous with $f_{n+1} \le f_n$ for all n. Show that if $f_n \to 0$ pointwise on K, then $f_n \to 0$ uniformly on K.

Solution.

Step 1. Fix $\epsilon > 0$; $U_n = \{x \in K : f_n(x) < \epsilon\} = V_n \cap K$ with V_n open (continuity), $U_n \subset U_{n+1}$ ($f_{n+1} \le f_n$), and $K \subset \bigcup_n V_n$ ($f_n \to 0$ pointwise).

Step 2. By compactness $K \subset V_{n_1} \cup \cdots \cup V_{n_r}$, so $K \subset U_N$, $N = \max_j n_j$.

Step 3. For $n \ge N$, $0 \le f_n \le f_N < \epsilon$ on K: $\|f_n\|_{\sup} \le \epsilon$.

No hypothesis can be dropped: x^n on $[0, 1)$ (not compact), $nx(1-x)^n$ on $[0, 1]$ (not monotone; maximum $\to e^{-1}$). ■

Exercise 6.6.12 (A zero from an integral)

Let $f(x)=\sum_{n\ge 0}a_n x^n$ have radius of convergence greater than 1, with real a_n satisfying

$$0<a_0<-\sum_{k=1}^{\infty}\frac{a_{2k}}{2k+1}.$$

Show that f has at least one zero in $(-1,1)$.

Solution.

Step 1. The series converges uniformly on $[-1,1]$ (Theorem 6.12), with $\sum_k|a_{2k}|<\infty$; integrate termwise (Proposition 6.5), using $\int_{-1}^{1}x^n dx=0$ (n odd), $\frac{2}{n+1}$ (n even):

$$\int_{-1}^{1}f(x)\,dx=2\Big(a_0+\sum_{k=1}^{\infty}\frac{a_{2k}}{2k+1}\Big)<0.$$

Step 2. If f had no zero in $(-1,1)$, then $f(0)=a_0>0$ and the intermediate value theorem give $f>0$ there, so $\int_{-1}^{1}f\ge 0$, contradicting Step 1. ■

Exercise 6.6.13 (Square of the alternating harmonic series)

Let $a_n=\frac{(-1)^{n+1}}{n+1}$ $(n\ge 0)$. Show that the Cauchy product of $\sum a_n$ with itself is

$$2\sum_{m=1}^{\infty}\frac{(-1)^{m+1}}{m+1}\Big(1+\frac{1}{2}+\cdots+\frac{1}{m}\Big),$$

that this series converges, and that its sum is $(\log 2)^2$.

Solution.

Step 1. (The product.) Write $H_m=1+\frac{1}{2}+\cdots+\frac{1}{m}$. For $n\ge 0$,

$$c_n=\sum_{k=0}^{n}a_k a_{n-k}=(-1)^n\sum_{k=0}^{n}\frac{1}{(k+1)(n+1-k)}.$$

By $\frac{1}{(k+1)(n+1-k)}=\frac{1}{n+2}\big(\frac{1}{k+1}+\frac{1}{n+1-k}\big)$ the sum is $\frac{2H_{n+1}}{n+2}$; with $m=n+1$, $c_{m-1}=2(-1)^{m+1}\frac{H_m}{m+1}$.

Step 2. (Convergence.) Let $b_m=\frac{H_m}{m+1}$. Since $H_{m+1}=H_m+\frac{1}{m+1}$,

$$b_m-b_{m+1}=\frac{(m+2)H_m-(m+1)H_{m+1}}{(m+1)(m+2)}=\frac{H_m-1}{(m+1)(m+2)}\ge 0.$$

$H_m\le 1+\log m$ ($\frac{1}{k}\le\int_{k-1}^{k}\frac{dx}{x}$), so $b_m\to 0$; $\sum(-1)^{m+1}b_m$ converges (Corollary 6.19).

Step 3. (The sum.) $\sum a_n=-\log 2$ (Example 6.4) converges only conditionally, so Mertens (Proposition 6.14) fails to apply; but all three series converge, and Corollary 6.16 gives $(\log 2)^2$. ■

Exercise 6.6.14 (Tauber's theorem)

Let $f(x)=\sum_{n\ge 0}a_n x^n$ converge on $(-1,1)$, and suppose $\lim_{x\to 1^-}f(x)=L$ and $\lim_n na_n=0$. Let $s_n=\sum_{k\le n}a_k$.
(a) Show that $\lim_{n\to\infty}(s_n-f(1-\frac{1}{n}))=0$.
(b) Show that $\sum_{n\ge 0}a_n=L$.

Solution.

(a) Let $x_n = 1 - \frac{1}{n}$ $(n \geq 2)$. Then

$$s_n - f(x_n) = \sum_{k \leq n} a_k (1 - x_n^k) - \sum_{k > n} a_k x_n^k.$$

Step 1. For $0 \leq x < 1$, $1 - x^k = (1 - x)(1 + x + \cdots + x^{k-1}) \leq k(1 - x)$. Hence

$$\Big| \sum_{k \leq n} a_k (1 - x_n^k) \Big| \leq \frac{1}{n} \sum_{k=0}^{n} k|a_k| = \frac{n+1}{n} \cdot \frac{1}{n+1} \sum_{k=0}^{n} k|a_k| \to 0,$$

as Cesàro means of the null sequence $k|a_k|$ tend to 0 (Lemma 9.12).

Step 2. $\epsilon_n = \sup_{k>n} k|a_k| \to 0$ and $|a_k| < \epsilon_n / n$ for $k > n$, so

$$\Big| \sum_{k > n} a_k x_n^k \Big| \leq \frac{\epsilon_n}{n} \sum_{k \geq 0} x_n^k = \frac{\epsilon_n}{n} \cdot \frac{1}{1 - x_n} = \epsilon_n \to 0.$$

Steps 1 and 2 give (a).

(b) $s_n = f(x_n) + (s_n - f(x_n)) \to L$ by (a).

Without $na_n \to 0$ this fails: $a_n = (-1)^n$ gives $f(x) = \frac{1}{1+x} \to \frac{1}{2}$, yet $\sum a_n$ diverges. ■